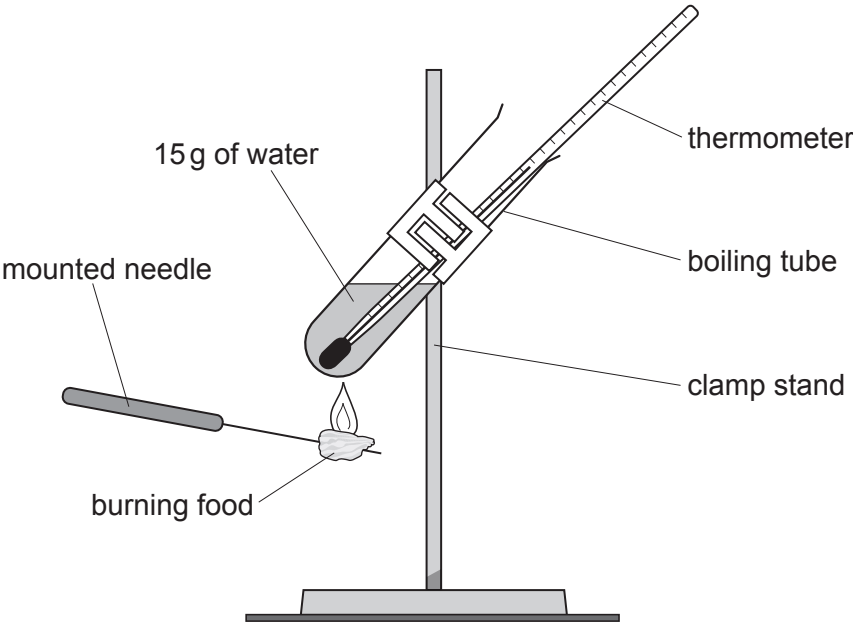


5. Some students investigated the energy content of food.



Their results are shown in the table below.

Food	Mass of food burnt (g)	Initial water temperature (°C)	Final water temperature (°C)	Temperature increase (°C)
bread	2.5	18	23	5
cheese	4.0	17	31	14
cornflakes	3.0	17	29	12
oatie biscuit	5.0	18	25	7

(a) Use the equation

$$\text{energy released} = 15 \times 4.2 \times \text{temperature increase}$$

to calculate the energy released by the bread.

[2]

energy released = J



Examiner
only

- (b) The students were asked to use their results to calculate the energy provided by a snack consisting of 15 g of oatie biscuits and 20 g of cheese. Complete the tables below to work out their answer.

[3]

Energy from 5 g of oatie biscuits (J)	Energy from 15 g of oatie biscuits (J)
441	

Energy from 4 g of cheese (J)	Energy from 20 g of cheese (J)
882	

Space for working

total energy from snack = J

5

